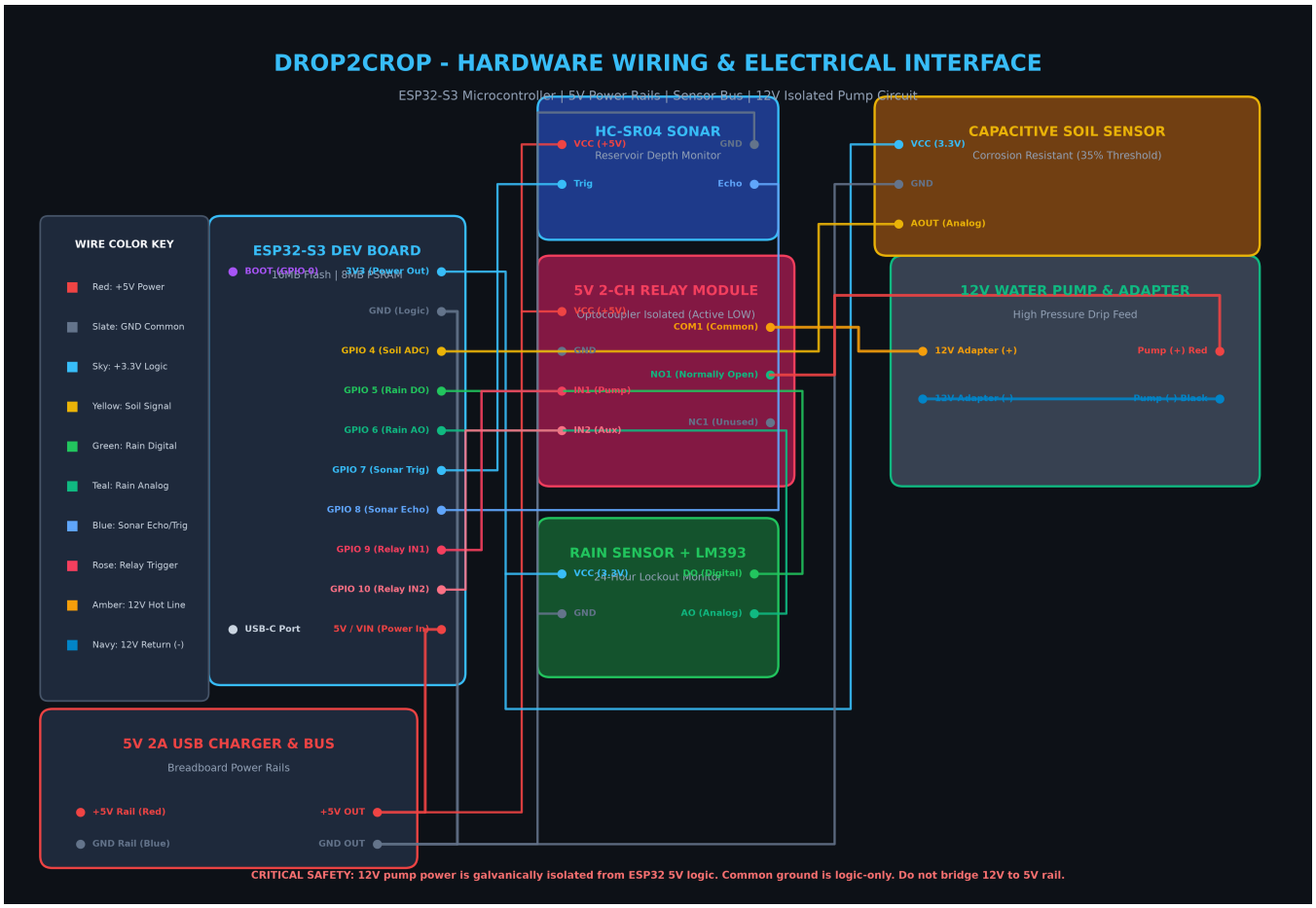
# Drop2Crop — Hardware Wiring Diagram & Manual

## Complete Electrical Pinout, Power Bus Distribution & Relay Circuitry

## 1. System Master Wiring Schematic

The diagram below illustrates all physical wire connections between the ESP32-S3, dual-channel relay module, HC-SR04 ultrasonic distance sensor, capacitive soil moisture sensor, rain detector, breadboard power rails, and 12V water pump.



## 2. Complete ESP32-S3 Pinout Connection Table

| Module / Component | Component Pin | ESP32-S3 Pin / Bus | Voltage | Signal Description |
|---|---|---|---|---|
| **Capacitive Soil Sensor** | VCC | **ESP32 3V3 Pin** | 3.3V DC | Clean 3.3V power (prevents ADC drift) |
| | GND | **Breadboard GND Rail** | 0V | Common Ground reference |
| | AOUT | **GPIO 4 (ADC1_CH3)** | 0 - 3.3V | Analog soil moisture signal (1350-3150) |
| **Rain Sensor (LM393)** | VCC | **ESP32 3V3 Pin** | 3.3V DC | Logic supply (safe for ESP32 inputs) |
| | GND | **Breadboard GND Rail** | 0V | Common Ground reference |
| | DO | **GPIO 5** | 3.3V Logic | Digital rain trigger (LOW = Rain) |
| | AO | **GPIO 6 (ADC1_CH5)** | 0 - 3.3V | Analog precipitation density |
| **HC-SR04 Ultrasonic** | VCC | **Breadboard +5V Rail** | 5.0V DC | 5V power for sonar transmitter |
| | GND | **Breadboard GND Rail** | 0V | Common Ground reference |
| | TRIG | **GPIO 7** | 3.3V Logic | 10µs ultrasonic trigger pulse |
| | ECHO | **GPIO 8** | Input | Echo pulse return width |
| **5V 2-Channel Relay** | VCC | **Breadboard +5V Rail** | 5.0V DC | Relay coil power supply |
| | GND | **Breadboard GND Rail** | 0V | Common Ground reference |
| | IN1 | **GPIO 9** | Active LOW | Relay 1 trigger: 12V Water Pump |
| | IN2 | **GPIO 10** | Active LOW | Relay 2 trigger: Aux / Fertigation |
| **Onboard Hardware** | BOOT Button | **GPIO 0** | Active LOW | Hold 3s to trigger WiFi Captive Portal |

## 3. Power Distribution Architecture (5V Charger + 12V Pump)

To ensure electrical stability and prevent micro-resets caused by inductive motor surges, the Drop2Crop system separates logic power from actuator power:

- **5V Mobile Phone Charger:** Powers the ESP32-S3 either via its USB-C port or by connecting the charger's +5V and GND wires to the breadboard rails, then linking the rail to the ESP32 **5V / VIN** and **GND** pins.
- **Breadboard Power Rails:** Distribute clean +5V to the Relay Module VCC and HC-SR04 VCC. Provide common Ground to all sensors and modules.
- **3.3V Logic Rail:** The ESP32's onboard low-dropout (LDO) regulator provides 3.3V from its **3V3 pin**. Connect this exclusively to the Capacitive Soil Moisture Sensor and the Rain Sensor comparator board.
- **12V High-Power Pump Isolation:** The 12V Water Pump is powered by an independent 12V Universal Adapter. The 12V (+) wire passes through the Relay 1 **COM (Common)** terminal and exits from the **NO (Normally Open)** terminal to the pump (+) lead. The 12V (-) wire connects directly to the pump (-) lead. **DO NOT connect the 12V power adapter to the breadboard or ESP32 pins.**

**ELECTRICAL SAFETY WARNING: Always ensure the 12V adapter is unplugged while inserting jumper wires into the relay screw terminals. The relay module provides optoisolation to protect the ESP32 from back-EMF spikes.**